

ASCII MASTER FLOW – PANEL 1/12: Polar climate world distribution

ICE-CAP -> Greenland and high-latitude highlands, permanent snow cover

TUNDRA -> coastal Greenland, N. Canada, Alaska, Eurasian Arctic seaboard

SOUTHERN POLE -> Antarctica: continent under ice, no indigenous population

NORTHERN POLE -> Arctic: ocean basin surrounded by continental land masses

MUST REMEMBER: Polar climates separate tundra ET, where the warmest month is above 0°C but below 10°C , from ice-cap EF below 0°C year-round; low sun angle, high albedo, polar night/day, permafrost and short food webs structure Arctic and Antarctic systems.

ASCII MASTER FLOW – PANEL 2/12: Temperature-precipitation profile

WARMEST MONTH -> seldom exceeds about 10 degrees C (June/July)

MONTHS ABOVE FREEZING -> no more than about four per year

PRECIPITATION -> low, mostly snow; polar climate = cold desert

BLIZZARDS -> wind redistributing existing snow, not heavy snowfall events

ASCII MASTER FLOW – PANEL 3/12: Tundra vegetation belt

PERMAFROST -> prevents deep rooting, impedes drainage

SHORT GROWING SEASON -> warmest month below 10 degrees C

STUNTED PLANTS -> dwarf willows, birches, alders in sheltered spots

LICHEN GROUND COVER -> reindeer moss provides grazing for reindeer herds

ASCII MASTER FLOW – PANEL 4/12: Arctic-Antarctic asymmetry

ARCTIC -> ocean surrounded by continents; mainly floating sea ice

ANTARCTIC -> continent surrounded by ocean; vast grounded ice sheet

WARMING -> Arctic amplification much faster; Antarctic more complex

GOVERNANCE -> Arctic: sovereign states + forum | Antarctic: treaty regime

ASCII MASTER FLOW – PANEL 5/12: Ice-albedo feedback loop

SEA ICE AND SNOW RETREAT -> bright surface replaced by dark ocean/ground

ALBEDO DROPS -> far more solar radiation absorbed instead of reflected

SURFACE WARMING -> further ice and snow retreat (self-reinforcing)

REINFORCING FACTORS -> thin polar atmosphere, ocean heat escape, poleward transport

ASCII MASTER FLOW – PANEL 6/12: Permafrost carbon feedback

WARMING -> thaws frozen ground containing ancient organic matter

DRY CONDITIONS -> decomposition releases carbon dioxide

WATERLOGGED CONDITIONS -> decomposition releases methane (stronger GHG)

FEEDBACK -> released GHGs cause further warming, further thaw

ASCII MASTER FLOW – PANEL 7/12: Sea-level grounded-ice rule

FLOATING SEA ICE MELTS -> sea level does NOT rise (already displacing water)

GROUNDING ICE MELTS -> mass transferred to ocean, sea level RISES

ICE SHELVES LOST -> removes buttressing, accelerates grounded-ice discharge

EXAM RULE -> always distinguish floating from grounded in any polar answer

ASCII MASTER FLOW – PANEL 8/12: Arctic shipping constraints

ADVANTAGE -> routes shorter than Suez/Panama between NE Asia and NW Europe
MOBILE ICE -> unpredictable, hazardous despite retreat trend
RESCUE AND CHARTING -> thin capacity, incomplete surveys
NEAR-TERM TRAFFIC -> resource export and destination, not transit container

ASCII MASTER FLOW – PANEL 9/12: India cold desert Ladakh

RAIN-SHADOW -> Greater Himalaya blocks SW monsoon, Trans-Himalaya stays arid

HIGH ALTITUDE -> intense insolation by day, sharp radiative cooling by night

WINTER PRECIPITATION -> light snow from western disturbances

BIOME CLASS -> cold desert, same as Sierra Nevada (USA) and Andes (Argentina)

ASCII MASTER FLOW – PANEL 10/12: India polar stations

DAKSHIN GANGOTRI (1983) -> first station, now decommissioned/buried in ice

MAITRI (1989) -> Schirmacher Oasis, Antarctica; operational

BHARATI (2012) -> Larsemann Hills, Antarctica; operational

HIMADRI (2008) -> Ny-Alesund, Svalbard; Arctic; HIMANSH -> Spiti, HP

CLOSE DISTINCTION: Arctic is mainly an ocean surrounded by continents while Antarctica is a high continent surrounded by ocean; sea ice differs from land ice, tundra from ice cap, and polar desert refers to low precipitation rather than absence of frozen water.

ASCII MASTER FLOW – PANEL 11/12: India polar legal framework

ANTARCTIC TREATY 1959 -> India acceded 1983; consultative party

INDIAN ANTARCTIC ACT 2022 -> domestic legislation for polar activities

ARCTIC POLICY 2022 -> six pillars; observer in Arctic forum

RESEARCH FOCUS -> teleconnections: Arctic ice loss to Indian monsoon disruption

EVIDENCE LIMIT: Amplification, sea-ice loss, ice-sheet mass balance and governance claims
require hemisphere, variable, baseline and date; Ladakh/cold deserts are useful process
comparisons but not polar climates.

ASCII MASTER FLOW – PANEL 12/12: Polar climate answer spine

DEFINE -> cold desert north of Arctic Circle; ice-cap vs tundra subdivisions

DISTINGUISH -> Arctic ocean vs Antarctic continent; floating vs grounded ice

EXPLAIN -> ice-albedo feedback, permafrost carbon, four transmission channels

QUALIFY -> India's Ladakh analogue, polar research, 2021 PYQ four-channel spine